Knowledge Representation for Model Fair-Lending Validation

B.1 Submission Form

Field	Entry
Name	Stuti Srivastava
ID No	2025AG05747
Industry / Domain	Banking and financial services / fair-lending validation
Organisation Type	Financial institution
[Instructor prompt]	Identify a real workplace process where decisions are governed by predefined rules.

B.2 Anchor (1 mark)

Workplace artifact

I work as a risk analyst. The selected artifact is a Model Fair-Lending Validation Decision Checklist and its supporting review record. This functional title represents the workplace record used to assess consumer impact, document the analyst's risk category and rationale, select suitable evidence and record legal guidance.

Why it matters

Models can influence lending decisions, customer treatment and financial risk. Incorrect classification or incomplete validation can result in unsuitable model use, missed testing, inconsistent reviews or unexplained exceptions. Fair-lending review helps assess potential discriminatory treatment or outcomes under applicable laws, principally the Equal Credit Opportunity Act (ECOA) and, for relevant residential real-estate transactions, the Fair Housing Act (FHA). These are distinct laws within the U.S. fair-lending framework. OCC: Fair Lending

How the process flows

1. **Submit the candidate tool:** The model owner provides its business purpose, intended use, methodology, inputs, outputs and supporting documentation.
2. **Classify the tool:** The reviewer applies the organisation's classification criteria to determine whether it is a model and which governance requirements apply.
3. **Assess risk and consumer impact:** The reviewer considers the tool's significance, intended use and influence on customer decisions to determine the appropriate review scope.
4. **Perform validation:** The risk analyst evaluates the model's conceptual soundness, data suitability, implementation and performance. Outcomes analysis and ongoing monitoring assess whether it performs as intended. These are central elements of the historical SR 11-7 validation framework. SR 11-7 attachment
5. **Conduct applicable fair-lending testing:** Select suitable tests under internal policy, considering the use case, available data and feasibility. Document limitations and escalate any proposed alternative testing approach.
6. **Record findings and review:** Document analyst conclusions, legal/compliance advice, remediation requirements and the authorised governance decision.

Who is involved

The Model Owner provides use-case, input and supporting evidence. The Risk Analyst assesses materiality, records the final fair-lending risk category, selects and interprets tests, and documents conclusions. The Legal Reviewer provides legal guidance, handles unresolved materiality and reviews vendor affirmation.

Where the facts come from

Facts come from the model inventory, classification checklist, development documentation, business-use descriptions, data dictionaries, validation datasets, model outputs, observed outcomes and monitoring reports. Internal policies, analyst assessments and recorded legal/governance decisions provide the rules and approval evidence.

Why a rule-based knowledge base fit

The checklist contains repeatable conditional decisions that can be expressed as logical rules. A knowledge base can consistently recommend classifications, review requirements and testing paths while explaining which facts support each conclusion. Missing evidence or ambiguous assessments can be routed for human review. Organisation-specific risk categories and testing routes comes from organisation's internal policy.

Scope note: The risk categories, testing routes and correlation threshold in this assignment represent the supplied workplace scenario. They are not presented as universal regulatory requirements. All cases and coefficients are synthetic.

SR 11-7 was superseded by SR 26-2 on April 17, 2026. For the assignment purpose, SR 11-7 is used as it's a historical foundation. Link for revised guidance. Federal Reserve: SR 26-2

B.3 Apply (2 marks)

B.3.1 Propositions, predicates and logical conventions

Every assessment has an explicit M(x) registration fact. The KB contains 24 business/inference-rule schemas and 82 explicit ground facts for seven synthetic model assessments. Each proposition describes one fixed model/use assessment. The corresponding unary predicate P(x) describes the same property for any assessment x. Constants a01–a07 identify the cases in B.3.3.

∀x means “for every assessment”; ∧ means AND, ∨ means OR, ¬ means explicit NOT, and → means implication. Every rule has equivalent propositional and universally quantified predicate forms. Existential quantifiers are unnecessary for these unary rules.

Open-world interpretation: an unrecorded fact is UNKNOWN, not false. Explicit ¬AMB means materiality is recorded as resolved; ¬DF means the selected DIA approach is recorded as infeasible. Suggested risk categories PH/PM/PL are separate from the analyst’s final recorded categories H/MD/L/O4.

Code	Meaning	Code	Meaning
M	Registered model/use assessment	CI	Consumer impact affirmatively assessed
FR	Fair-lending review required	AMB	Consumer-impact materiality currently unresolved
LR	Legal review required	HOLD	Assessment held for review
AL	Application lending	AF	Application-stage fraud
DM	Direct marketing	MI	Material consumer impact
TF	Transaction-stage fraud	SV	Servicing
IM	Indirect marketing	DP	Deposit-related use
SR	Semantic offer search	AO	All eligible offers accessible
NM	Offer-placement effect assessed as non-material	SA	Analyst-recorded servicing assessment
PH	Suggested High risk	PM	Suggested Medium risk
PL	Suggested Low risk	CH	Analyst selects High
CM	Analyst selects Medium	CL	Analyst selects Low
CO	Analyst selects Out of Scope	DEC	Analyst risk decision recorded
RAT	Decision rationale recorded	LGR	Applicable legal guidance recorded
H	Final analyst-recorded risk 1 High	MD	Final analyst-recorded risk 2 Medium
L	Final analyst-recorded risk 3 Low	NI	No consumer impact affirmatively assessed

B.3 Apply - Vocabulary continued

Code	Meaning	Code	Meaning
O4	Final analyst-recorded risk 4 Out of Scope for fair-lending validation	IS	Input screening required
T	Tabular inputs or score data	IA	Attributes accessible for CA
CA	Correlation analysis recommended	X	Text inputs used
TXREQ	Text-control decision required	ADVCH	Adversarial testing selected
REDCH	PII redaction selected	ADV	Adversarial testing recommended
RED	PII redaction recommended	TXDONE	All chosen text controls completed
TXACC	Analyst acceptance of text-control evidence recorded	TXTOK	Text-control review recorded as satisfied
DF	Selected DIA approach assessed as feasible	DIA	DIA assessment recommended
ALT	Alternative evidence review required	VEND	Vendor model
VREQ	Vendor assurance requested through legal	VA	Vendor affirmation/testing statement received
LV	Legal acceptance of vendor evidence recorded	PILOT	Pilot-testing evidence available
SCOREPERF	Score-performance evidence available	VE	Vendor-evidence review record established; not proof of no bias
CADONE	All declared feature/protected-variable correlation results complete	CALOW	Every required absolute correlation is below 0.30
CAACC	Analyst acceptance of CA evidence recorded	CAOK	CA screening criterion and acceptance recorded
CABREACH	At least one required |r| is at least 0.30	AR	Analyst attention or resolution required
DIAACC	Analyst/legal interpretation of DIA evidence accepted	DIAREC	Accepted DIA review record established

Evidence conventions used in the rules

The confirmed correlation criterion is strictly $|r| < 0.30$ for every required feature/protected-variable comparison. Values of +0.30 or -0.30 fail that criterion. Complete coverage and analyst acceptance are separate facts; a low coefficient alone is not a fairness conclusion.

DIA is Disparate Impact Analysis. The workplace's statistical review can include an F-test, with hypothesis and interpretation determined by the analyst. A two-variance F-test assesses variance differences; it does not by itself establish absence of lending bias. No universal significance level or automatic fairness pass is assumed.

The risk levels and 0.30 criterion describe the student's reported workplace practice. They are not presented as universal legal thresholds. All case facts and numeric coefficients below are illustrative.

B.3 Apply - Business rules and logical forms

B.3.2 Consumer impact and risk assessment

Each numbered row is a business rule and a KB inference-rule schema. PL denotes propositional logic; FOL denotes predicate logic. The same symbols have the definitions given in B.3.1.

Business rule	Equivalent propositional and predicate forms
R01 A registered model with consumer impact requires fair-lending review.	PL: $M \land CI \rightarrow FR$ FOL: $\forall x\,[M(x) \land CI(x) \rightarrow FR(x)]$
R02 Unresolved materiality for a model requires legal review and a hold on its assessment.	PL: $M \land AMB \rightarrow LR \land HOLD$ FOL: $\forall x\,[M(x) \land AMB(x) \rightarrow LR(x) \land HOLD(x)]$
R03 Resolved application lending or application-stage fraud suggests High risk.	PL: $FR \land \neg AMB \land (AL \lor AF) \rightarrow PH$ FOL: $\forall x\,[FR(x) \land \neg AMB(x) \land (AL(x) \lor AF(x)) \rightarrow PH(x)]$
R04 Resolved material direct marketing or transaction-stage fraud suggests Medium risk.	PL: $FR \land \neg AMB \land ((DM \land MI) \lor TF) \rightarrow PM$ FOL: $\forall x\,[FR(x) \land \neg AMB(x) \land ((DM(x) \land MI(x)) \lor TF(x)) \rightarrow PM(x)]$
R05 Resolved servicing, indirect marketing or deposit use suggests Low risk.	PL: $FR \land \neg AMB \land (SV \lor IM \lor DP) \rightarrow PL$ FOL: $\forall x\,[FR(x) \land \neg AMB(x) \land (SV(x) \lor IM(x) \lor DP(x)) \rightarrow PL(x)]$
R06 Resolved non-material semantic offer search, with all offers accessible and an analyst-recorded servicing assessment, suggests Low risk.	PL: $FR \land \neg AMB \land SR \land AO \land NM \land SA \rightarrow PL$ FOL: $\forall x\,[FR(x) \land \neg AMB(x) \land SR(x) \land AO(x) \land NM(x) \land SA(x) \rightarrow PL(x)]$
R07 Record High risk from the analyst's selected category, decision, rationale and legal guidance after materiality is resolved.	PL: $FR \land \neg AMB \land CH \land DEC \land RAT \land LGR \rightarrow H$ FOL: $\forall x\,[FR(x) \land \neg AMB(x) \land CH(x) \land DEC(x) \land RAT(x) \land LGR(x) \rightarrow H(x)]$
R08 Record Medium risk from the corresponding analyst decision and evidence.	PL: $FR \land \neg AMB \land CM \land DEC \land RAT \land LGR \rightarrow MD$ FOL: $\forall x\,[FR(x) \land \neg AMB(x) \land CM(x) \land DEC(x) \land RAT(x) \land LGR(x) \rightarrow MD(x)]$

B.3 Apply - Business rules and logical forms

B.3.2 Risk records and input controls

Each numbered row is a business rule and a KB inference-rule schema. PL denotes propositional logic; FOL denotes predicate logic. The same symbols have the definitions given in B.3.1.

Business rule	Equivalent propositional and predicate forms
R09 Record Low risk from the corresponding analyst decision and evidence.	PL: FR ∧ ¬AMB ∧ CL ∧ DEC ∧ RAT ∧ LGR → L FOL: ∀x [FR(x) ∧ ¬AMB(x) ∧ CL(x) ∧ DEC(x) ∧ RAT(x) ∧ LGR(x) → L(x)]
R10 Record fair-lending Out of Scope for a model with an explicit no-consumer-impact assessment and a documented analyst decision.	PL: M ∧ NI ∧ ¬AMB ∧ CO ∧ DEC ∧ RAT ∧ LGR → O4 FOL: ∀x [M(x) ∧ NI(x) ∧ ¬AMB(x) ∧ CO(x) ∧ DEC(x) ∧ RAT(x) ∧ LGR(x) → O4(x)]
R11 Low or Medium risk requires input screening.	PL: (L ∨ MD) → IS FOL: ∀x [(L(x) ∨ MD(x)) → IS(x)]
R12 Tabular input screening with accessible attributes recommends correlation analysis.	PL: IS ∧ T ∧ IA → CA FOL: ∀x [IS(x) ∧ T(x) ∧ IA(x) → CA(x)]
R13 Consumer-impact text inputs require a text-control decision.	PL: FR ∧ X → TXREQ FOL: ∀x [FR(x) ∧ X(x) → TXREQ(x)]
R14 A text-control decision selecting adversarial testing recommends that test.	PL: TXREQ ∧ ADVCH → ADV FOL: ∀x [TXREQ(x) ∧ ADVCH(x) → ADV(x)]
R15 A text-control decision selecting PII redaction recommends that control.	PL: TXREQ ∧ REDCH → RED FOL: ∀x [TXREQ(x) ∧ REDCH(x) → RED(x)]
R16 Chosen text controls must be completed and accepted by the analyst before the text review is recorded as satisfied.	PL: (ADV ∨ RED) ∧ TXDONE ∧ TXACC → TXTOK FOL: ∀x [(ADV(x) ∨ RED(x)) ∧ TXDONE(x) ∧ TXACC(x) → TXTOK(x)]

B.3 Apply - Business rules and logical forms

B.3.2 DIA, vendor evidence and interpretation

Each numbered row is a business rule and a KB inference-rule schema. PL denotes propositional logic; FOL denotes predicate logic. The same symbols have the definitions given in B.3.1.

Business rule	Equivalent propositional and predicate forms
R17 High risk and recorded feasibility recommend the selected DIA approach.	PL: H ∧ DF → DIA FOL: ∀x [H(x) ∧ DF(x) → DIA(x)]
R18 High risk with recorded DIA infeasibility requires alternative evidence review.	PL: H ∧ ¬DF → ALT FOL: ∀x [H(x) ∧ ¬DF(x) → ALT(x)]
R19 A consumer-impact vendor model with unavailable attributes and an infeasible DIA approach requires vendor assurance through legal.	PL: FR ∧ VEND ∧ ¬IA ∧ ¬DF → VREQ FOL: ∀x [FR(x) ∧ VEND(x) ∧ ¬IA(x) ∧ ¬DF(x) → VREQ(x)]
R20 Vendor affirmation accepted by legal, together with pilot and score-performance evidence, establishes the vendor-evidence record.	PL: VREQ ∧ VA ∧ LV ∧ PILOT ∧ SCOREPERF → VE FOL: ∀x [VREQ(x) ∧ VA(x) ∧ LV(x) ∧ PILOT(x) ∧ SCOREPERF(x) → VE(x)]
R21 Complete CA results, all required |r| values below 0.30 and recorded analyst acceptance establish the input-screening result.	PL: CA ∧ CADONE ∧ CALOW ∧ CAACC → CAOK FOL: ∀x [CA(x) ∧ CADONE(x) ∧ CALOW(x) ∧ CAACC(x) → CAOK(x)]
R22 Any |r| value at or above 0.30 requires analyst attention.	PL: CABREACH → AR FOL: ∀x [CABREACH(x) → AR(x)]
R23 A recommended DIA assessment with documented analyst/legal acceptance establishes an accepted DIA review record.	PL: DIA ∧ DIAACC → DIAREC FOL: ∀x [DIA(x) ∧ DIAACC(x) → DIAREC(x)]
R24 Overlapping suggested risk categories require analyst resolution with documented rationale.	PL: ((PH ∧ PM) ∨ (PH ∧ PL) ∨ (PM ∧ PL)) → AR FOL: ∀x [((PH(x) ∧ PM(x)) ∨ (PH(x) ∧ PL(x)) ∨ (PM(x) ∧ PL(x))) → AR(x)]

B.3 Apply - Knowledge Base

B.3.3 Facts and inference rules

KB = F ∪ R. F contains the 82 ground literals below; R contains R01–R24. Positive and explicitly negated predicates each count as one fact. Rule-derived conclusions are excluded from the initial count. All data and recorded decisions in these examples are synthetic.

Constant / case	Explicit ground facts	N
a01 Underwrite_A	AL(a01), ¬AMB(a01), CH(a01), CI(a01), DEC(a01), DF(a01), DIAACC(a01), IA(a01), LGR(a01), M(a01), RAT(a01), T(a01)	12
a02 Marketing_B	¬AMB(a02), CAACC(a02), CADONE(a02), CALOW(a02), CI(a02), CM(a02), DEC(a02), DM(a02), IA(a02), LGR(a02), M(a02), MI(a02), RAT(a02), T(a02)	14
a03 OfferSearch_C	ADVCH(a03), ¬AMB(a03), AO(a03), CI(a03), CL(a03), DEC(a03), LGR(a03), M(a03), NM(a03), RAT(a03), SA(a03), SR(a03), TXACC(a03), TXDONE(a03), X(a03)	15
a04 StressTest_D	¬AMB(a04), CO(a04), DEC(a04), LGR(a04), M(a04), NI(a04), RAT(a04)	7
a05 VendorScore_E	AF(a05), ¬AMB(a05), CH(a05), CI(a05), DEC(a05), ¬DF(a05), ¬IA(a05), LGR(a05), LV(a05), M(a05), PILOT(a05), RAT(a05), SCOREPERF(a05), T(a05), VA(a05), VEND(a05)	16
a06 PendingOffer_F	AMB(a06), AO(a06), CI(a06), M(a06), SR(a06), X(a06)	6
a07 ServicingText_G	¬AMB(a07), CI(a07), CL(a07), DEC(a07), LGR(a07), M(a07), RAT(a07), REDCH(a07), SV(a07), TXACC(a07), TXDONE(a07), X(a07)	12

Fact count: 12 + 14 + 15 + 7 + 16 + 6 + 12 = 82.
Inference-rule count: 24 named schemas (R01–R24).

Correlation evidence for a02

Feature	Race	Age	Gender
income	0.12	-0.25	0.10
utilisation	0.08	0.20	-0.15

All six declared comparisons are present and the maximum absolute value is 0.25. CADONE(a02) and CALOW(a02) represent that evidence. CAACC(a02) separately records analyst acceptance. Race/age/gender are the illustrative protected-variable labels; production coding and group contrasts require a documented protocol.

B.3 Apply - Five workplace queries

Methods follow the reference: Q1 and Q4 use forward chaining; Q2 uses backward chaining; Q3 uses a variable query and forward explanations; Q5 includes ground resolution refutation. Missing facts remain unknown unless an explicit opposite is recorded.

Q1. What fair-lending review applies to Underwrite_A?

Step / rule	Premises	Conclusion
1 · R01	M(a01), CI(a01)	FR(a01)
2 · R03	FR(a01), ¬AMB(a01), AL(a01)	PH(a01)
3 · R07	FR(a01), ¬AMB(a01), CH(a01), DEC(a01), RAT(a01), LGR(a01)	H(a01)
4 · R17	H(a01), DF(a01)	DIA(a01)
5 · R23	DIA(a01), DIAACC(a01)	DIAREC(a01)

Answer: High risk is recorded by the analyst; the feasible DIA route is recommended, and the synthetic analyst/legal acceptance establishes an accepted DIA review record. No conclusion that a single statistical result proves absence of discrimination is inferred.

Q2. Does Marketing_B meet the input-screening criterion?

Method: backward chaining starts with CAOK(a02) and resolves its supporting subgoals.

Step / rule	Premises	Conclusion
1 · R21	Goal CAOK(a02): needs CA, CADONE, CALOW and CAACC. The last three are recorded evidence facts.	Prove CA(a02)
2 · R12	CA(a02) needs IS(a02), T(a02) and IA(a02). T and IA are asserted.	Prove IS(a02)
3 · R11, R08	IS follows from MD. MD needs FR plus ¬AMB, CM, DEC, RAT and LGR, all recorded.	Prove FR(a02)
4 · R01	FR(a02) needs M(a02) and CI(a02), both asserted. All subgoals are supported.	CAOK(a02) proved

Answer: Yes. All six declared absolute correlations are below 0.30 and the analyst acceptance is recorded. This satisfies the stated input-screening criterion; it is not a conclusion that every possible fair-lending risk has been eliminated.

Q3. How are text-input models and unresolved materiality handled?

The variable query FR(x) ∧ X(x) returns a03, a06 and a07. The cases below explain the supported review actions.

Case	Premises and rules	Result
a03 OfferSearch_C	R01 derives FR. Resolved materiality, SR/AO/NM/SA suggest Low through R06. CL/DEC/RAT/LGR establish L through R09. X and ADVCH select ADV through R13/R14; TXDONE and TXACC establish TXTOK through R16.	Recorded Low; adversarial text-control review satisfied.
a06 PendingOffer_F	CI establishes FR by R01. AMB establishes LR and HOLD by R02. No resolved-materiality or final-category decision facts establish H, MD, L or O4.	Legal review and hold; final category remains unknown.

Answer: a03 has a recorded Low category and accepted adversarial controls; a07 uses accepted PII-redaction controls. The unresolved a06 assessment is referred to legal and held. Text input alone does not determine the final category.

For the separate servicing-text case a07, R09 establishes L; R13/R15 select PII redaction from REDCH; R16 recognises the completed and accepted text-control review. Both adversarial testing and redaction are represented without assuming that selecting a control means it has been performed.

Q4. Does StressTest_D require fair-lending validation?

Step / rule	Premises	Conclusion

Step / rule	Premises	Conclusion
1 · R10	M(a04), NI(a04), ¬AMB(a04), CO(a04), DEC(a04), RAT(a04), LGR(a04)	O4(a04)
2 · R01	No CI(a04) fact supports this rule; NI(a04) explicitly records no consumer impact.	FR(a04) not derived
3 · R11, R17	No Low/Medium or High final risk supports these test paths.	IS, CA and DIA not derived

Answer: The analyst-recorded outcome is risk 4: Out of Scope for this fair-lending validation process, based on the explicit no-consumer-impact assessment. The result follows from the recorded assessment, not from the "StressTest" name.

Q5. What evidence is used for an opaque vendor score?

Step / rule	Premises	Conclusion
1 · R01, R03	M(a05), CI(a05); FR(a05), ¬AMB(a05), AF(a05)	FR(a05), PH(a05)
2 · R07	FR(a05), ¬AMB(a05), CH(a05), DEC(a05), RAT(a05), LGR(a05)	H(a05)
3 · R18	H(a05), ¬DF(a05)	ALT(a05)
4 · R19	FR(a05), VEND(a05), ¬IA(a05), ¬DF(a05)	VREQ(a05)
5 · R20	VREQ(a05), VA(a05), LV(a05), PILOT(a05), SCOREPERF(a05)	VE(a05)

Answer: The selected DIA approach is recorded as infeasible and the attributes needed for CA are explicitly unavailable. Legal obtains/reviews vendor affirmation, while pilot and score-performance evidence support the review. The KB establishes a vendor-evidence record rather than proposing a CA test without attributes.

Vendor affirmation is a statement about the vendor's testing and is recorded as evidence; it does not independently prove absence of bias. Likewise, infeasibility of the selected score/probability approach is not a claim that every possible group comparison or alternative review is mathematically impossible.

Resolution refutation for VE(a05)

Ground R20 at x = a05: ¬VREQ(a05) ∨ ¬VA(a05) ∨ ¬LV(a05) ∨ ¬PILOT(a05) ∨ ¬SCOREPERF(a05) ∨ VE(a05). Add the negated goal ¬VE(a05). VREQ(a05) is established by R19 in the preceding trace.

Resolve with	Remaining resolvent
Negated goal ¬VE(a05)	¬VREQ(a05) ∨ ¬VA(a05) ∨ ¬LV(a05) ∨ ¬PILOT(a05) ∨ ¬SCOREPERF(a05)
VREQ(a05)	¬VA(a05) ∨ ¬LV(a05) ∨ ¬PILOT(a05) ∨ ¬SCOREPERF(a05)
VA(a05)	¬LV(a05) ∨ ¬PILOT(a05) ∨ ¬SCOREPERF(a05)
LV(a05)	¬PILOT(a05) ∨ ¬SCOREPERF(a05)
PILOT(a05)	¬SCOREPERF(a05)
SCOREPERF(a05)	□ - empty clause

The empty clause shows that the negated evidence-record goal contradicts the KB, so KB ⊨ VE(a05). This is a conclusion about a reviewed evidence record, not proof of absence of bias.

Registered model scope

Every assessment begins with an explicit M(x) fact. Model registration is the starting scope assumption for the KB; the fair-lending rules then assess consumer impact, risk and appropriate evidence.

B.4 Analyse (1 mark)

How Predicate Logic improves expressiveness and reasoning

Workplace need	Propositional Logic	Predicate Logic
Reuse across assessments	CI → FR concerns one fixed assessment; each additional model needs separately grounded symbols.	∀x [CI(x) → FR(x)] expresses one rule for every model/use assessment.
Trace a decision to its case	The candidate's identity is implicit in the symbol.	H(a01) and H(a05) identify separate recorded decisions and their supporting facts.
Ask inventory-wide questions	A fixed finite list can be enumerated with many propositions.	FR(x) returns every supported assessment requiring review; H(x) returns a01 and a05.
Distinguish evidence from conclusions	Separate symbols are needed for each fixed case.	VA(x), VE(x) and DIAREC(x) preserve the distinction between received evidence, reviewed evidence and an accepted assessment.

Predicate Logic provides compact, reusable representation and explicit candidate identity. It does not automatically improve data quality or make a business assumption correct. For a fixed finite inventory, Propositional Logic can still represent the same grounded decisions.

Effectiveness for the selected workplace problem

The KB gives consistent, explainable answers for consumer-impact scope, materiality escalation, final risk records, tabular screening, text controls and vendor evidence. It separates suggested categories from analyst-recorded decisions and keeps missing information visible. The queries show how changing one premise-such as feasibility or resolved materiality-changes the supported review path.

Missing knowledge and limitations affecting decisions

Real testing requires an approved correlation measure, categorical encodings/group contrasts, complete feature inventory, test populations, sample requirements and data-quality checks. These are not determined by a universal 0.30 limit. The limit is a screening criterion reported for this workplace; correlation does not establish causation or a legal finding.

F-test hypotheses, significance levels, distributional assumptions and practical interpretation are case-specific analyst decisions. A non-significant variance comparison does not demonstrate equal outcomes or absence of discrimination. Vendor affirmation may have scope, age and independence limitations that the analyst/legal record must evaluate.

The academic KB uses unary snapshot facts. It does not track individual reviewers, attribute/group relationships, evidence dates or changing use cases directly. Contradictory inputs must be rejected or resolved; missing facts remain unknown. A real implementation needs source verification, versioned evidence and reassessment when relevant information changes.

B.5 Reflect (1 mark)

Practical constraints in organisational deployment

Constraint	Effect on fair-lending validation
Incomplete knowledge	Missing input attributes, group/outcome data or materiality evidence can prevent a justified test recommendation. Gaps need explicit review rather than a default assumption of safety.
Evolving business rules	Product uses, internal risk definitions and legal guidance may change. Rules and decision records need an identified owner, version and effective date.
Privacy and permitted data use	Protected-class and personally identifiable information require approved access, purpose and controls. The academic examples use synthetic facts and coefficients.
Vendor transparency	Proprietary features, training data or score transformations may be unavailable. Document vendor evidence, pilot results and legal review while retaining the limitations.
Scalability and technology	A real inventory has many models, features and evidence records. Reliable identifiers, efficient rule evaluation and integration with review systems are necessary.
Expert judgment	Materiality, statistical interpretation and evidence adequacy need accountable analyst/legal decisions. A rule recommendation does not replace those decisions.

Future Scope of improvement

Introduce a version-controlled fair-lending evidence and decision record linking each KB fact to its source, assessment date, analyst rationale and applicable legal guidance.

Even correct rules can produce an unsuitable conclusion from stale, incomplete or unsupported evidence. The record should preserve materiality decisions, final risk rationale, test populations and protocols, correlation coverage, DIA interpretation, text-control results and vendor-evidence limitations. It should make clear who accepted the conclusion and which evidence supported it.

The examples demonstrate why this matters: the unresolved offer-search case needs a genuine legal/materiality decision; the marketing screen needs complete, correctly encoded comparisons; and the vendor case needs evidence whose scope and limitations can be examined. Adding more rules would not compensate for missing evidence.

Appendix A - Python implementation

python Fair_Lending_KB.py



Fair_Lending_KB.py

Code Run Output: forward chaining for Query 1

```
Problems  Output  Debug Console  Terminal  Ports                                                    Python

PS C:\Mtech\Sem2\ACI\Assignments> & 'C:\Users\stuti\anaconda3\python.exe' 'c:/Mtech/Sem2/ACI/Assignments/Fair_Lending_KB.py'
Initial facts: 82 | Rule schemas: 24
Derived facts: 36 | Total facts: 118

============================================================
Q1: Underwrite_A — forward chaining
PH(a01) [R03]
  FR(a01) [R01]
    M(a01) [FACT]
    CI(a01) [FACT]
  ~AMB(a01) [FACT]
  AL(a01) [FACT]
H(a01) [R07]
  FR(a01) [R01]
    M(a01) [FACT]
    CI(a01) [FACT]
  ~AMB(a01) [FACT]
  CH(a01) [FACT]
  DEC(a01) [FACT]
  RAT(a01) [FACT]
  LGR(a01) [FACT]
DIA(a01) [R17]
  H(a01) [R07]
    FR(a01) [R01]
      M(a01) [FACT]
      CI(a01) [FACT]
    ~AMB(a01) [FACT]
    CH(a01) [FACT]
    DEC(a01) [FACT]
    RAT(a01) [FACT]
    LGR(a01) [FACT]
  DF(a01) [FACT]
DIAREC(a01) [R23]
  DIA(a01) [R17]
    H(a01) [R07]
      FR(a01) [R01]
        M(a01) [FACT]
        CI(a01) [FACT]
      ~AMB(a01) [FACT]
      CH(a01) [FACT]
      DEC(a01) [FACT]
      RAT(a01) [FACT]
      LGR(a01) [FACT]
    DF(a01) [FACT]
  DIAACC(a01) [FACT]
Answer: High risk; DIA recommended; accepted DIA review record established.
```

Captured output: backward chaining for Query 2

```
Q2: Marketing_B — backward chaining
CAOK(a02) [R21]
  CA(a02) [R12]
    IS(a02) [R11]
      MD(a02) [R08]
        FR(a02) [R01]
          M(a02) [FACT]
          CI(a02) [FACT]
        ~AMB(a02) [FACT]
        CM(a02) [FACT]
        DEC(a02) [FACT]
        RAT(a02) [FACT]
        LGR(a02) [FACT]
    T(a02) [FACT]
    IA(a02) [FACT]
  CADONE(a02) [FACT]
  CALOW(a02) [FACT]
  CAACC(a02) [FACT]
Input-screening criterion: TRUE
All six declared comparisons are complete; maximum |r| = 0.25; analyst acceptance is recorded.
```

Figure A1. Actual program output rendered from backward_query2.txt. The tree shows CAOK(a02), supporting rules and the asserted premises.

Captured output: resolution refutation for Query 5

```
============================================================
Q5: VendorScore_E — evidence and resolution refutation
H(a05) [R07]
  FR(a05) [R01]
    M(a05) [FACT]
    CI(a05) [FACT]
  ~AMB(a05) [FACT]
  CH(a05) [FACT]
  DEC(a05) [FACT]
  RAT(a05) [FACT]
  LGR(a05) [FACT]
ALT(a05) [R18]
  H(a05) [R07]
    FR(a05) [R01]
      M(a05) [FACT]
      CI(a05) [FACT]
    ~AMB(a05) [FACT]
    CH(a05) [FACT]
    DEC(a05) [FACT]
    RAT(a05) [FACT]
    LGR(a05) [FACT]
  ~DF(a05) [FACT]
VREQ(a05) [R19]
  FR(a05) [R01]
    M(a05) [FACT]
    CI(a05) [FACT]
  VEND(a05) [FACT]
  ~IA(a05) [FACT]
  ~DF(a05) [FACT]
VE(a05) [R20]
  VREQ(a05) [R19]
    FR(a05) [R01]
      M(a05) [FACT]
      CI(a05) [FACT]
    VEND(a05) [FACT]
    ~IA(a05) [FACT]
    ~DF(a05) [FACT]
  VA(a05) [FACT]
  LV(a05) [FACT]
  PILOT(a05) [FACT]
  SCOREPERF(a05) [FACT]

Resolution refutation:
Ground R20: VE(a05) OR ~LV(a05) OR ~PILOT(a05) OR ~SCOREPERF(a05) OR ~VA(a05) OR ~VREQ(a05)
Resolve with ~VE(a05): ~LV(a05) OR ~PILOT(a05) OR ~SCOREPERF(a05) OR ~VA(a05) OR ~VREQ(a05)
Resolve with VREQ(a05): ~LV(a05) OR ~PILOT(a05) OR ~SCOREPERF(a05) OR ~VA(a05)
Resolve with VA(a05): ~LV(a05) OR ~PILOT(a05) OR ~SCOREPERF(a05)
Resolve with LV(a05): ~PILOT(a05) OR ~SCOREPERF(a05)
Resolve with PILOT(a05): ~SCOREPERF(a05)
Resolve with SCOREPERF(a05): EMPTY CLAUSE
Answer: Reviewed vendor-evidence record established.

Evidence records and screening criteria do not prove absence of discrimination.
```

Figure A2. Actual program output rendered from resolution_query5.txt. Each resolvent records its parent clause numbers and pivot; the trace ends with the empty clause.